

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2013-14

Paper: CSB-502 — Net Centric Computing

Time: Three hours

Maximum Marks: 70

Instructions: ANSWER FIVE QUESTIONS INCLUDING QUESTION NO. 1 WHICH IS COMPULSORY.

Question 1.

- (a) What is meant by flow control? [2]
- (b) What are the fields of the UDP header? [3]
- (c) Distinguish between bus topology and ring topology. [3]
- (d) What is the count-to-infinity problem? [3]
- (e) What are the disadvantages of the DES algorithm? [3]

Question 2.

- (a) Explain the four basic network topologies, and cite an advantage of each type. [5]
- (b) What are the responsibilities of the network layer in the Internet model? [5]
- (c) Describe the multiplexing techniques used to combine digital signals. [4]

Question 3.

- (a) Explain in detail, with examples, the various error detecting methods. [7]
- (b) Suppose that we model "A Simplex Protocol for a Noisy Channel" using the finite state machine model. How many states exist for the communication channel? How many states exist for the complete system? [7]

Question 4.

- (a) What is internetworking? Discuss the various global addressing schemes and the issues in forwarding IP packets. [7]
- (b) Explain why routing is very important in networking. Illustrate any one of the routing algorithms used in networks. [7]

Question 5.

- (a) Differentiate between connectionless and connection-oriented services at the Transport Layer. Draw the header format of TCP and explain the purpose of each field. [7]

- (b) What is the difference between open-loop congestion control and closed-loop congestion control? Name the policies that can prevent congestion. [7]

Question 6.

- (a) What is the difference between a primary server and a secondary server? What are the two main categories of DNS messages? [4]
- (b) Write short notes on:
- (i) WWW
 - (ii) Browser
 - (iii) URL

[6]

- (c) What is the mechanism of cookies? [4]

Question 7.

- (a) Describe Simple Network Management Protocol (SNMP) and explain the role of MIB (Management Information Base). [7]
- (b) In asymmetric key cryptography, how do you think two persons can establish two pairs of keys between themselves? Explain with an example. [7]